

Smart Healthcare: How IoT is Reshaping the Industry

Healthcare is being transformed by the Internet of Things (IoT), and becoming more effective, data-driven, and patient-centred. We take a look at the transformative impact of IoT on the healthcare industry, outlining its potential to improve patient treatment and increase overall healthcare.



The healthcare sector has seen a significant surge in the adoption of Internet of Things (IoT) devices during 2023. Based on a report published by *MarketsandMarkets*, the worldwide market for IoT in the healthcare sector is anticipated to reach \$188 billion by 2025. This growth is expected to be achieved through a compound yearly growth rate (CAGR) of 29.9% from 2020 to 2025.

Figure 1 shows IoT's growth trajectory in the healthcare business.

Key applications of IoT in healthcare

IoT has diverse applications in healthcare, driving improvements in patient care, clinical processes, and operational efficiency (Figure 2).

Patient monitoring: Through the Internet of Things, doctors can remotely monitor their patients' vital signs, the progress of long-term illnesses, and post-surgery recovery. They can identify potential issues early, and deliver timely interventions, reducing the likelihood of hospital readmissions.

Clinical decision support: IoT applications provide real-time patient data, empowering healthcare professionals to make well-informed decisions, thus improving patient care and reducing the occurrence of medical errors.

Device management: IoT is used to manage the status of medical equipment within healthcare facilities. This minimises equipment loss, optimises asset utilisation, and ensures the availability of critical resources.

Predictive maintenance: IoT aids in predicting when medical devices require maintenance or replacement, thereby reducing downtime and ensuring the continuous availability of equipment for patient care.

Research and innovation: IoT-enabled devices and wearables are employed in clinical trials to collect participant data, significantly improving the efficiency and accuracy of medical research, and aiding in drug development.

IoT benefits in healthcare

There are various benefits of integrating IoT in healthcare (Figure 3).

Enhanced patient engagement: With the support of the Internet of Things (IoT), medical portals